

Moments, levers and gears

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1	State what is required to cause an object to rotate.	A force that does not act through the pivot or centre of mass.	☐
2	Name the turning effect of a force.	A moment	☐
3 ★	State which distance is used when calculating the moment of a force.	The perpendicular distance from the pivot to the line of action of the force	☐
4	The clockwise and anticlockwise moments about a pivot are equal. State what this tells you about the object.	It is balanced.	☐
5 ★	Explain how a lever can reduce the force needed to turn something.	Pushing further from the pivot gives the same moment with a smaller force.	☐
6	State the function of a lever.	To transmit the rotational effect of a force.	☐
7	State the function of a simple gear system.	To transmit the rotational effect of a force.	☐
8	State the equation linking moment, force and distance.	moment = force × perpendicular distance from the pivot ($M = F d$)	☐
9	State the standard unit of the moment of a force.	Newton-metre (N m)	☐
10 ★	State the principle of moments for a balanced object.	Total clockwise moment about the pivot = total anticlockwise moment.	☐
11	A small gear drives a larger gear. State the effect on the speed of rotation and on the moment.	The larger gear turns more slowly but with a bigger moment.	☐